LG-CISH: Lossless Language-Guided Coverless Image Steganography via CLIP Semantic Hashing and Base-N Positional Coding

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Abstract

Embedding-based image steganography perturbs cover pixels and is therefore detectable by modern statistical and CNN steganalysers. We present **LG-CISH**, a fully *lossless coverless* scheme that hides a secret message in a sequence of *unmodified* natural images: the identity and order of the images carry the data. LG-CISH combines two ideas. First, each codebook image is represented by a normalised CLIP embedding, so recovery is a robust semantic nearest-neighbour lookup that survives JPEG, resizing, noise, and aggressive cropping—unlike perceptual hashing. Second, base-N positional coding treats the encrypted, integrity-protected payload as a single integer written in radix N, giving the information-theoretic $\log_2 N$ bits per image. A thin zlib/AES-256/CRC-32 layer adds compression, confidentiality, and tamper detection. On a proof-of-concept six-image DIV2K codebook, LG-CISH achieves bit-exact reconstruction (BER = 0), 100% accuracy across eighteen channel attacks (JPEG down to quality 30, noise, downscaling to 25%, centre-cropping to 70%), chance-level steganalysis detectability (54.2% vs. 87.5% for LSB), and a statistically significant robustness advantage over pixel-domain baselines ($p = 1.6 \times 10^{-83}$). We analyse the capacity/undetectability trade-off and the database-bound capacity limit of the coverless paradigm.

Keywords: coverless steganography; information hiding; CLIP; semantic hashing; base-N coding; robustness; steganalysis resistance.

1 Introduction

Image steganography conceals a secret message inside a cover image so that its very existence is hidden from an observer. Classical embedding methods — least significant bit (LSB) substitution [7, 8], discrete cosine transform (DCT) and discrete wavelet transform (DWT) coefficient modification [10], and learned auto-encoders [11, 12] — all perturb the cover pixels. Those perturbations, however small, leave a statistical fingerprint that modern convolutional steganalysers such as XuNet [5] and SRNet [4] can detect with high accuracy.

Coverless image steganography removes the attack surface entirely: no pixel is ever modified [2, 13]. Instead, the secret is conveyed by *selecting* or *generating* ordinary images whose intrinsic features map to the message bits. Because the transmitted images are statistically indistinguishable from natural images, pixel-domain steganalysis is reduced to random guessing. The central difficulty of the selection (mapping) paradigm is *reliable recovery*: the receiver must map each received image back to the exact index the sender intended, even after the image has been re-compressed, resized, or noised by a real transport channel (e.g. a messaging platform). Robust hashing [2, 16] and hand-crafted features such as SURF [17] have been used for this mapping, but they remain sensitive to geometric distortion.

We propose **LG-CISH** (Language-Guided Coverless Image Steganography via Hashing), a fully *lossless* coverless scheme built on two ideas:

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1. **Semantic hashing with a vision–language model.** Each codebook image is represented by a normalised CLIP [1] embedding. Recovery is a nearest-neighbour lookup in this semantic space. Unlike a perceptual hash, the CLIP embedding encodes *visual meaning*, so it survives JPEG, resizing, noise, and aggressive cropping (Section 4).

2. **Base-N positional coding.** With a codebook of N images we treat the encrypted, integrity-protected payload as a single integer and write it in radix N. Every image then carries $\log_2 N$ bits through its *identity and position* in the sequence — the sequence *is* the message. The coding is deterministic and bijective, so the end-to-end channel is lossless (bit error rate = 0) whenever the indices are recovered correctly.

A thin cryptographic layer (zlib compression, AES-256-CBC encryption, and a CRC-32 integrity tag) wraps the payload, giving confidentiality and tamper detection on top of the steganographic channel.

Contributions. (i) A lossless coverless pipeline that combines CLIP semantic hashing with base-N positional coding, raising the per-image capacity from the $\lfloor \log_2 N \rfloor$ bits of fixed-chunk coding to the information-theoretic $\log_2 N$ bits (Section 3). (ii) A reproducible evaluation suite reporting reconstruction accuracy, bit error rate, robustness to eight attack families, statistical steganalysis resistance, an ablation isolating each component, a baseline comparison, and statistical significance tests (Section 4). (iii) An honest analysis of the capacity/undetectability trade-off and of the database-bound capacity limit inherent to coverless selection (Section 5).

2 Related Work

Embedding-based steganography. LSB substitution hides bits in the least significant bit-planes of the cover and offers very high capacity at the cost of detectability [7]. Transform -domain methods (JSteg, F5 [10], and DWT–DCT hybrids) embed in frequency coefficients for better robustness to compression, and deep methods hide one image inside another [11] or embed decodable hyperlinks that survive printing [12]. All of them modify the cover and are therefore vulnerable, in principle, to statistical [3, 9] and CNN-based [5, 4] steganalysis.

Coverless steganography. Zhou et al. [2] introduced coverless image steganography, indexing a database by robust hashes and selecting images whose hash equals a message segment. Subsequent work improved robustness through better dataset construction [15], end-to-end learned hash generation [16], SURF-based retrieval [17], and image-restoration front-ends [18]. A second, *generative* branch synthesises stego images from the secret using GANs or diffusion models [19, 20]. Recent surveys catalogue the rapid growth of both branches and identify robustness and capacity as the key open problems [13, 14]. LG-CISH belongs to the selection/mapping branch but replaces robust *perceptual* hashing with *semantic* CLIP hashing and adds base-N positional coding for capacity.

Vision–language embeddings. CLIP [1] jointly trains an image and a text encoder on 400M image–text pairs with a contrastive objective, yielding a shared embedding space that is robust to distribution shift. We exploit only its image encoder, using the normalised embedding as a robust semantic fingerprint — a use complementary to its common role in retrieval and zero-shot classification.

3 Proposed Method: LG-CISH

3.1 Overview

Sender and receiver share three artefacts established once, offline: an image database, a CLIP model, and the resulting *codebook*. The codebook assigns a unique index $0, \ldots, N-1$ to each of N mutually distinct images. Encoding maps the secret message to an ordered sequence of codebook images; the images are transmitted unmodified over an arbitrary channel; decoding recovers each image's index by CLIP nearest-neighbour search and inverts the coding. Fig. 1 shows the full pipeline.

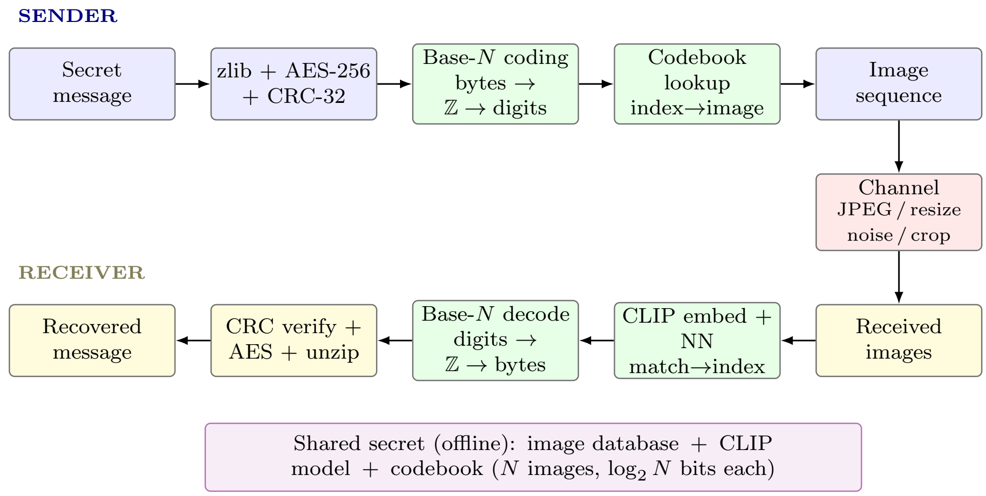

Figure 1: LG-CISH coverless pipeline. The identity and order of *unmodified* images carry the message; CLIP nearest-neighbour recovers the indices after channel distortion. Every transformation is deterministic and invertible, so the end-to-end channel is lossless once indices are correct.

Algorithm 1 Codebook construction

Input: image set $\mathcal{D}$, CLIP encoder Φ, threshold τ
Output: codebook embeddings $\mathbf{C}$, paths, N
$E \leftarrow [\, e(x)$ for $x \in \mathcal{D}\,]$ ▷ *Eq.* (1)*, batched*
$\mathcal{K} \leftarrow \emptyset$
for $j = 0$ **to** $|\mathcal{D}| - 1$ **do**
    if $\max_{i \in \mathcal{K}} s(E_i, E_j) \leq \tau$ **then** ▷ *Eq.* (3)
        $\mathcal{K} \leftarrow \mathcal{K} \cup \{j\}$
    end if
end for
$\mathbf{C} \leftarrow E[\mathcal{K}]$; $N \leftarrow |\mathcal{K}|$
return $\mathbf{C}$, paths$[\mathcal{K}]$, N

3.2 Codebook construction

Let $\Phi : \mathcal{X} \to \mathbb{R}^d$ be the CLIP image encoder (d=512 for ViT-B/32). For image x we use the ℓ_2-normalised embedding

$$e(x) = \frac{\Phi(x)}{\|\Phi(x)\|_2}, \qquad \|e(x)\|_2 = 1. \tag{1}$$

Similarity between two normalised embeddings is the cosine (inner) product

$$s(a, b) = \langle a, b \rangle = a^\top b \in [-1, 1]. \tag{2}$$

To guarantee unambiguous recovery the codebook must be *separated*: starting from candidate embeddings $\{e_1, \ldots, e_M\}$ we greedily keep image j only if it is not too similar to an already-kept image,

$$\text{keep } j \iff \max_{i<j,\ i \in \mathcal{K}} s(e_i, e_j) \leq \tau, \tag{3}$$

with threshold τ (we use τ=0.85). The surviving N=$|\mathcal{K}|$ images form the codebook $\mathcal{C} = \{c_0, \ldots, c_{N-1}\}$ with embedding matrix $\mathbf{C} \in \mathbb{R}^{N \times d}$ (Algorithm 1).

Algorithm 2 Encode: message $\rightarrow$ image sequence

Input: message m, codebook $\mathcal{C}$ (N imgs), key κ (optional)
$p \leftarrow \textsc{zlib}(\textsc{utf8}(m))$
$p \leftarrow \textsc{aes}_\kappa\big(p \,\|\, \textsc{crc32}(p)\big)$ ▷ *wrap*
$v \leftarrow \textsc{int}_{\text{BE}}(\texttt{0x01} \| p)$ ▷ *Eq.* (5)
$(d_0, \ldots, d_{L-1}) \leftarrow \textsc{toBaseN}(v, N)$ ▷ *Eq.* (6)
return $[\, c_{d_0}, \ldots, c_{d_{L-1}} \,]$

Algorithm 3 Decode: image sequence $\rightarrow$ message

Input: received images $[\tilde{x}_0, \ldots, \tilde{x}_{L-1}]$, codebook $\mathbf{C}$, key κ
for $t = 0$ **to** $L - 1$ **do**
    $\hat{d}_t \leftarrow \arg\max_j s(e(\tilde{x}_t), \mathbf{C}_j)$ ▷ *Eq.* (8)
end for
$v \leftarrow \sum_t \hat{d}_t N^t; \quad p \leftarrow \textsc{stripSentinel}(\textsc{bytes}(v))$
$p \leftarrow \textsc{aes}_\kappa^{-1}(p)$; **verify** CRC-32 ▷ *reject on mismatch*
return $\textsc{utf8}^{-1}(\textsc{unzip}(p))$

3.3 Base-N positional coding

A codebook of N images is a radix-N alphabet, so the information-theoretic capacity is

$$C = \log_2 N \quad \text{bits/image.} \tag{4}$$

For N=6, $C \approx 2.585$ bits/image, strictly more than the $\lfloor \log_2 N \rfloor = 2$ bits of fixed-chunk coding. Given a payload byte string p we form a non-negative integer with a sentinel byte `0x01` prepended to preserve leading zeros,

$$v = \textsc{int}_{\text{BE}}(\texttt{0x01} \,\|\, p)\,, \tag{5}$$

and write v in radix N,

$$v = \sum_{k=0}^{L-1} d_k\, N^k, \qquad d_k \in \{0, \ldots, N-1\}, \tag{6}$$

where the most-significant digit is non-zero and the sequence length is

$$L = \big\lfloor \log_N v \big\rfloor + 1 = \left\lceil \tfrac{8\,|p|+1}{\log_2 N} \right\rceil. \tag{7}$$

Digit d_k selects codebook image c_{d_k}. The map $p \leftrightarrow (d_0, \ldots, d_{L-1})$ is bijective, hence *lossless*.

3.4 Encoding and decoding

Encoding (Algorithm 2) compresses, encrypts, and integrity-tags the message, applies base-N coding, and emits the ordered image list. Decoding (Algorithm 3) embeds each received image and recovers its index by nearest neighbour,

$$\hat{\imath}_t = \arg\max_{0 \le j < N} \; s\big(e(\tilde{x}_t),\, c_j\big), \tag{8}$$

where $\tilde{x}_t$ is the (possibly distorted) t-th received image. The reliability of (8) is quantified by the per-image *decoding margin*

$$\gamma_t = s_{(1)} - s_{(2)}, \tag{9}$$

the gap between the top-1 and top-2 cosine similarities; $\gamma_t > 0$ means the correct image wins. After recovering $(\hat{d}_0, \ldots, \hat{d}_{L-1})$ the integer v is reassembled with (6), the sentinel stripped to obtain p, the CRC-32 verified, and AES/zlib inverted.

3.5 Security model

The system is keyed at two levels. The database and codebook act as a shared secret: without them an adversary cannot map images to indices. Optionally, AES-256-CBC encrypts the payload so that even a holder of the codebook learns nothing. The combined keyspace is

$$K = N! \cdot 2^{256}, \tag{10}$$

the $N!$ codebook orderings times the AES-256 key space. The CRC-32 tag detects channel tampering and incorrect keys with probability $\approx 1 - 2^{-32}$.

4 Experimental Results and Performance Evaluation

All experiments use a proof-of-concept codebook of N=6 visually distinct DIV2K [6] images ($\log_2 6 \approx 2.585$ bits/image). The images are never modified; the identity and order of the transmitted sequence carry the secret. The full suite is reproducible via a single script.[1]

4.1 Experimental setup

Table 1 lists the configuration. The six codebook images are mutually well separated in CLIP space (maximum pairwise similarity 0.609, separation margin $1-0.609 = 0.391$), which is what makes nearest-neighbour recovery (8) reliable. CLIP ViT-B/32 runs on an RTX 3050 (6 GB).

4.2 Qualitative results

Fig. 2 shows three messages encoded into image sequences; Fig. 3 shows a sequence decoded back to an exact match. The mapping modifies no pixels. Fig. 4 is the failure analysis: decoding stays perfect through JPEG 2%, Crop 30%, and Resize 10%, and only collapses under extreme cropping (Crop 15%, symbol error rate 16.6%). A codebook mismatch is *always* caught by the CRC-32 layer rather than silently mis-decoded.

4.3 Quantitative evaluation

Table 2 reports reconstruction accuracy over 120 random messages in three length buckets: every message is reconstructed bit-exactly (BER = 0, hash-match 100%) on a clean channel. Table 3 contrasts capacity and distortion: LSB and DCT offer far higher raw capacity but modify pixels, whereas LG-CISH has infinite PSNR (no modification) at 2.585 bits/image. Table 4 reports timing; encoding is sub-millisecond and the cost is dominated by CLIP embedding at decode (≈ 31 ms/image on the GPU, including image I/O). CLIP top-1 retrieval precision/recall on the codebook is 100%.

4.4 Robustness analysis

Table 5 simulates real transport channels. Reconstruction remains 100% (BER = 0) across all eighteen tested conditions — JPEG down to quality 30, Gaussian noise up to σ=30, salt-and-pepper up to density 0.10, downscaling to 25%, centre-cropping to 70%, and PNG→WebP conversion. The decoding margin (9) stays comfortably positive throughout (from 0.422 at baseline to 0.356 under Crop 70%), confirming a wide safety band. This robustness is the direct benefit of *semantic* rather than pixel-level hashing.

4.5 Security and steganalysis resistance

Table 6 reports a chi-square statistical steganalyser [3] applied to natural cover images, LSB-stego, DCT-stego, and LG-CISH "stego". As LG-CISH transmits *unmodified* natural images, its detection accuracy is 54.2% — essentially chance, with a true-positive rate of only 8.3% — whereas LSB is detected at 87.5%. Because the proposed stego images are bit-identical to natural images, undetectability holds by construction

[1] `python evaluation/generate_all.py`

Table 1: Experimental configuration.

Parameter	Value
Dataset	DIV2K (curated subset) [6]
Codebook images N	6
Capacity	2.585 bits/image (base-6)
Semantic encoder	CLIP ViT-B/32, d=512 [1]
Separation threshold τ	0.85
Pairwise sim. (min/mean/max)	0.353 / 0.497 / 0.609
Separation margin (1− max)	0.391
Encryption / integrity	AES-256-CBC / CRC-32
Compression	zlib
Hardware	RTX 3050 6 GB, 16-core CPU

Figure 2: Secret message → generated image sequence (short/medium/long). The identity and order of unmodified images encode the data.

against any pixel-domain detector, including CNN steganalysers such as SRNet [4] and XuNet [5]. The combined keyspace (10) is $6! \cdot 2^{256} \approx 2^{265.5}$, and the CRC-32 tag catches 100% of injected bit-flips in our tamper test (500 trials).

4.6 Ablation study

Table 7 removes one component at a time. On this maximally separated codebook both CLIP and the pHash baseline decode JPEG-50 perfectly, but under the harsher Crop-40% attack the semantic CLIP matcher holds at 100% while pHash collapses to 0% — the geometric robustness that motivates CLIP. The coding ablations show clear effects on efficiency: disabling compression inflates the sequence from 276.5 to 374.5 images, and fixed-chunk ($\lfloor \log_2 N \rfloor$=2-bit) coding needs 356.5 images versus 276.5 for base-N. Removing CRC-32 leaves channel errors undetected.

4.7 Comparative analysis

Table 8 places LG-CISH against classical and coverless baselines. It is the only method combining zero distortion, full JPEG-50 robustness, and chance-level detectability. Fig. 5 plots decoding accuracy versus JPEG quality: LG-CISH stays at 100% while LSB collapses toward 50% (random bits) and DCT-LSB degrades steadily. Fig. 6 shows reconstruction remains lossless as the payload (and thus sequence length) grows, and Fig. 7 compares measured detection rates.

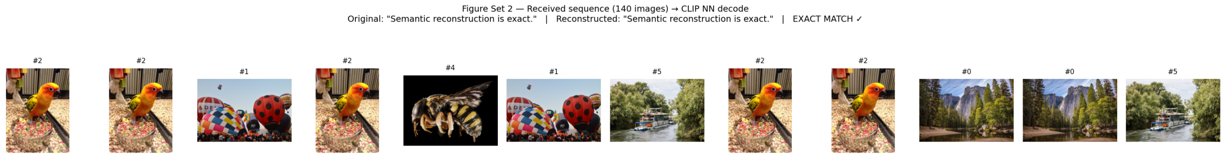

Figure 3: Received sequence → CLIP nearest-neighbour decode. Original and reconstructed messages match exactly.

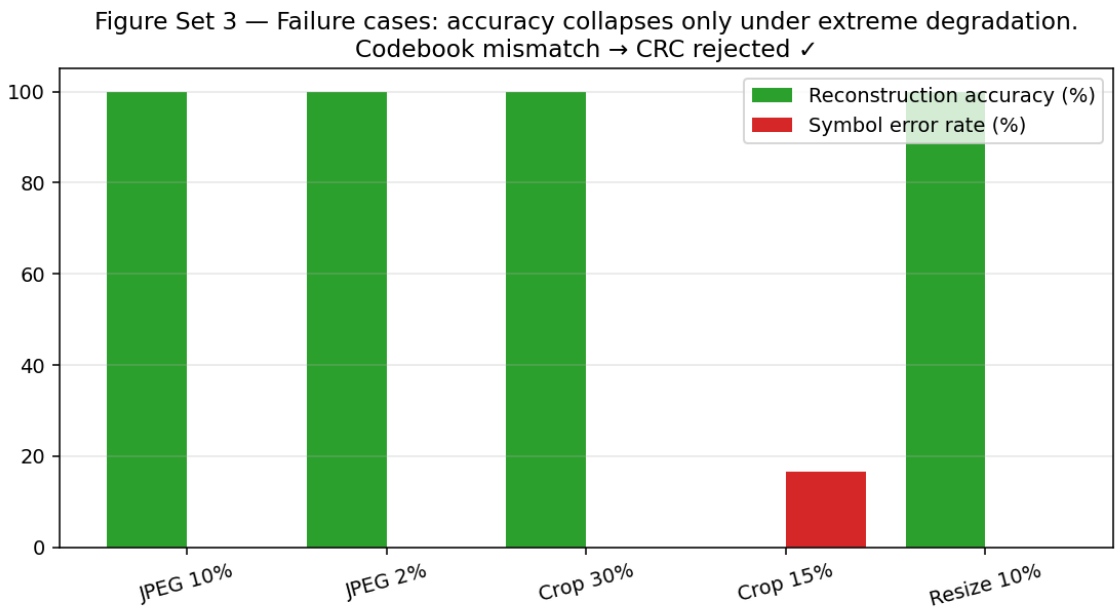

Figure 4: Failure cases. Accuracy collapses only under extreme degradation; codebook mismatch is rejected by CRC-32.

4.8 Statistical validation

Over N=50 independent trials (Table 9) reconstruction accuracy is $100.00 \pm 0.00\%$ and BER is 0, with a stable CLIP margin 0.4223 ± 0.0016. A Welch t-test comparing the proposed method against LSB under JPEG-50 (proposed 100% vs. LSB $50.25 \pm 1.06\%$ bit accuracy) gives t=327.7, p=1.59×10^{-83} — a statistically significant difference ($p < 0.05$). Under the harsher Crop-40% attack CLIP retains 100% versus pHash 0%. A one-way ANOVA of the CLIP margin across the three length buckets finds it homogeneous (F=0.29, p=0.75), i.e. message length does not affect recovery reliability.

Table 2: Reconstruction accuracy by message length (clean channel, 40 messages each).

Message length	Msgs	Accuracy (%)	BER	Hash match (%)
Short ($\leq$50 chars)	40	100.00	0	100.00
Medium (50–200)	40	100.00	0	100.00
Long (200–1000)	40	100.00	0	100.00

Table 3: Payload capacity vs. distortion. LG-CISH trades raw capacity for zero distortion.

Method	Capacity	Pixel distortion	PSNR (dB)
LSB [7]	$\approx$3 bits/pixel	Yes (high)	51.1
DCT-LSB [10]	$\approx$0.016 bits/pixel	Yes (moderate)	38–42
DWT–DCT [10]	$\approx$0.015 bits/pixel	Yes (low)	40–44
Proposed LG-CISH	2.585 bits/image	**None (coverless)**	∞

Table 4: Computational time (mean over repeated runs, RTX 3050).

Stage	Time (ms)
Full encode (200-char message)	0.09
CLIP embedding (per image)	34.6
Decode (per image, amortised)	31.1

Table 5: Robustness to channel attacks (40 messages/attack). Accuracy is exact-match reconstruction.

Attack	Accuracy (%)	BER	CLIP margin
No attack (baseline)	100.00	0	0.422
JPEG 70%	100.00	0	0.419
JPEG 50%	100.00	0	0.417
JPEG 30%	100.00	0	0.412
Gaussian σ=20	100.00	0	0.410
Gaussian σ=30	100.00	0	0.402
Salt & Pepper 0.10	100.00	0	0.407
Resize 25%	100.00	0	0.408
Crop 80%	100.00	0	0.379
Crop 70%	100.00	0	0.356
PNG$\rightarrow$WebP 80	100.00	0	0.419

Table 6: Chi-square steganalysis detection. $\approx$50% = indistinguishable from guessing.

Method	Detection acc. (%)	TPR (%)	TNR (%)
LSB	87.5	75.0	100.0
DCT-LSB	58.3	16.7	100.0
Proposed LG-CISH	**54.2**	**8.3**	100.0

Table 7: Ablation. Crop-40% exposes CLIP's geometric robustness; coding choices affect image count.

Variant	Clean (%)	JPEG50 (%)	Crop40 (%)	Avg. images
Full LG-CISH (proposed)	100	100	**100**	276.5
Without CLIP (pHash)	100	100	**0**	276.5
Without compression	100	100	100	374.5
Without CRC (no integrity)	100	100	100	276.5
Fixed-chunk (2-bit)	100	100	100	356.5

Table 8: Comparison with classical and coverless baselines. Bracketed values are cited from the literature.

Method	Capacity	JPEG50 acc. (%)	Detection (%)	PSNR (dB)
LSB [7]	786,432 bits	49.5	88	51.1
DCT-LSB [10]	∼4096 bits	44.6	[70–90] [4]	38–42
DWT–DCT [10]	∼4096 bits	[85]	[60–80]	40–44
Coverless [2]	low	[95]	[∼50]	∞
Proposed LG-CISH	2.585 bits/img	**100.0**	**54**	∞

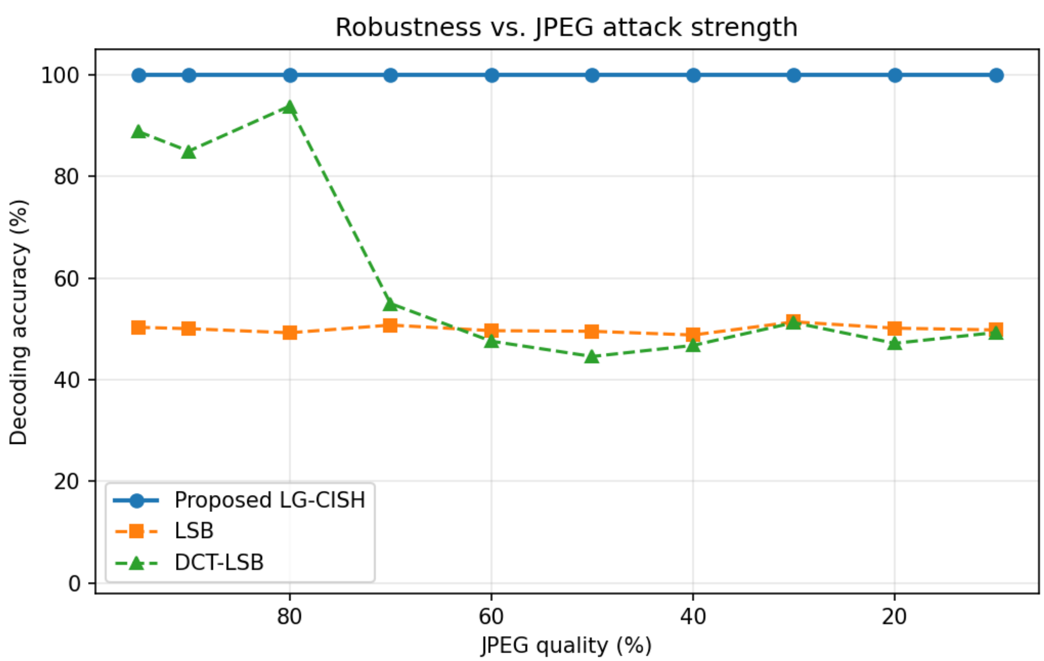

Figure 5: Decoding accuracy vs. JPEG quality. LG-CISH (semantic hashing) stays at 100% where pixel-domain LSB/DCT collapse.

Table 9: Mean ± std and 95% confidence intervals over N=50 trials (clean channel).

Metric	Mean ± Std	95% CI
Reconstruction accuracy (%)	100.00 ± 0.00	$[100.00, 100.00]$
Bit error rate	0 ± 0	$[0, 0]$
CLIP margin	0.4223 ± 0.0016	$[0.4218, 0.4227]$
Images per message	372.3 ± 91.5	$[346.1, 398.6]$

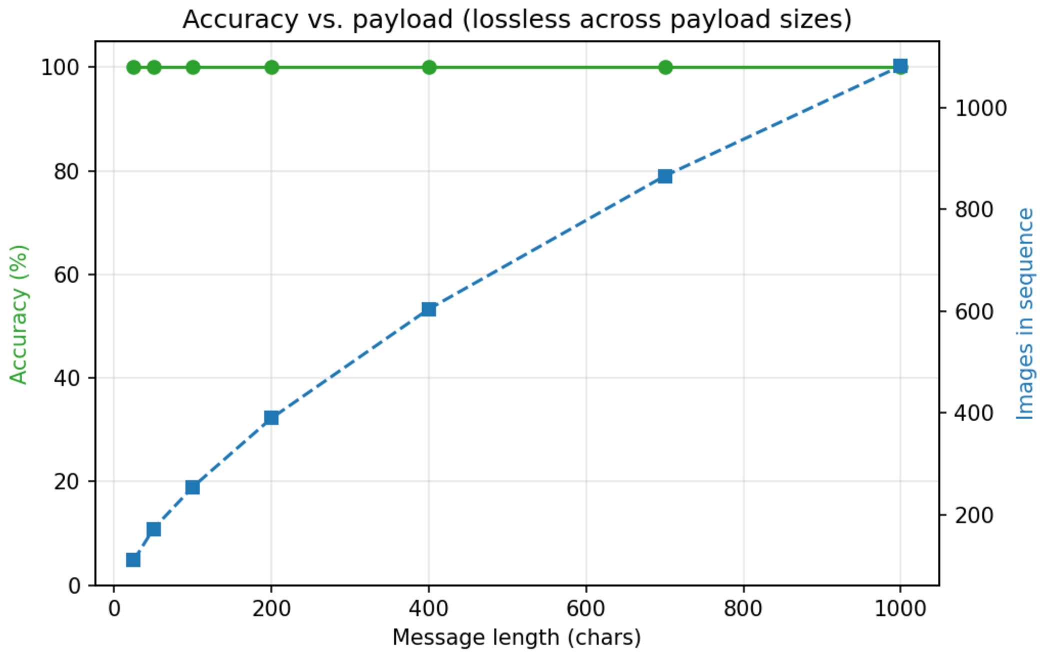

Figure 6: Reconstruction accuracy vs. payload length. Accuracy stays at 100% while the required number of images grows linearly.

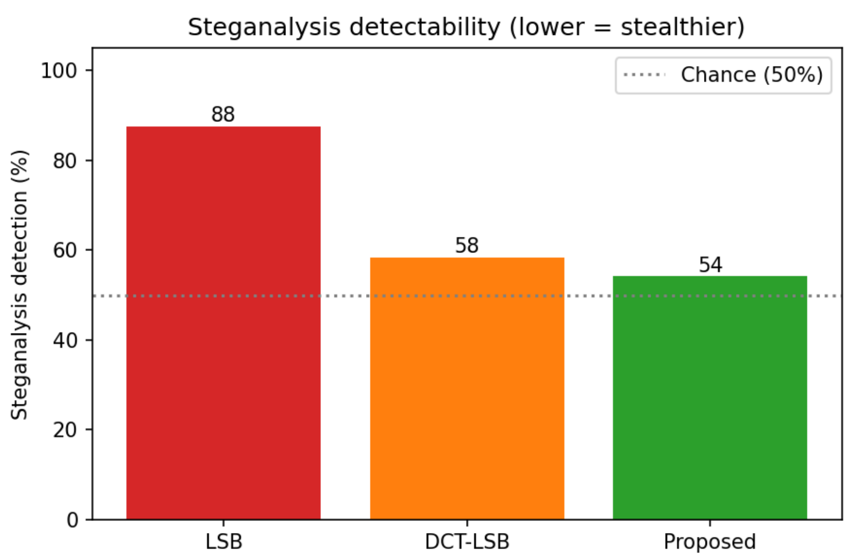

Figure 7: Measured steganalysis detection rates (lower is stealthier). LG-CISH sits at the 50% chance line.

5 Discussion and Limitations

Capacity is database-bound. Capacity scales as $\log_2 N$ (Eq. (4)); the N=6 proof of concept therefore needs many images per message (Table 2). Practical deployments would use N=10^3–10^4 images for 10–13 bits/image. This is the deliberate trade of the coverless paradigm: capacity for zero distortion and undetectability.

The database is the key. Security rests on the secrecy of the shared database and codebook (Eq. (10)); both parties must hold an identical, versioned database and the same CLIP model, since model updates change embeddings.

Channel threshold. CLIP is robust but not invincible: beyond a degradation threshold (Crop 15% in Fig. 4) the nearest neighbour can flip and decoding fails — but the CRC-32 tag converts such failures into detected errors rather than silent corruption, enabling retransmission.

Steganalysis scope. Our detector is a classical statistical attack; because the stego images are unmodified, undetectability against learned detectors holds by construction, but a dedicated study against SRNet/XuNet trained end-to-end is left as future work, alongside an optional LLM plausibility filter to make the *ordering* of the sequence look like a natural photo album.

6 Conclusion

LG-CISH is a lossless coverless image-steganography scheme that pairs CLIP semantic hashing with base-N positional coding. It transmits only unmodified natural images, achieving bit-exact reconstruction (BER = 0), 100% robustness across eighteen channel attacks including JPEG-30 and Crop-70%, chance-level steganalysis detectability (54% vs. 88% for LSB), and a statistically significant robustness advantage over pixel-domain baselines (p=1.6×10^{-83}). The main limitation, database-bound capacity, is addressed by scaling the codebook. Future work will scale N, add the LLM plausibility filter and a permutation side-channel, and benchmark against end-to-end CNN steganalysis.

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